

Sales Analysis Using Tableau - Project Documentation

1. Introduction

The Sales Analysis Dashboard project analyzes sales performance using Tableau data visualizations. The goal is to transform raw sales data into meaningful insights such as total sales, profit, product performance, regional distribution, and sales trends. The dashboard helps organizations make data-driven decisions.

2. Data Collection & Extraction

The dataset used for the project consists of four files: Customers.csv – customer information, Location.csv – city, state, region, postal code, Orders.csv – order date, sales, profit, quantity, Products.csv – product category and subcategory. The data was connected to Tableau using the Text File connection option and relationships were created between Orders, Customers, Products, and Location tables.

3. Data Preparation

Data preparation included cleaning missing values, checking duplicates, converting the order date field to date format, and validating relationships between tables. Calculated fields such as Total Sales, Total Profit, Profit Ratio, and Year were created for analysis.

4. Data Visualizations

Several visualizations were created: • KPI Total Sales • KPI Total Profit • KPI Total Quantity • Monthly Sales Trend • Subcategory Comparison Chart • Weekly Sales and Profit Trend Total visualizations created: 6

5. Dashboard Design

A responsive dashboard (size 1200 x 800) was created combining all visualizations. The dashboard includes KPI cards, comparison charts, and trend graphs. It is interactive, user-friendly, and includes filters for better exploration of the data.

6. Story Creation

A Tableau Story was created to explain insights step by step: Scene 1 – Sales Overview Scene 2 – Product Category Analysis Scene 3 – Regional Sales Analysis Scene 4 – Sales Trend Analysis Scene 5 – Final Insights Total scenes: 5

7. Performance Testing

Performance evaluation included filters, calculated fields, and optimized visualizations. Filters used include Year, Region, and Category. Around 4–6 calculated fields and 6 visualizations were used in the dashboard.

8. Web Integration with Flask

The Tableau dashboard was published to Tableau Public and embedded into a Flask web application using an iframe. Flask serves a webpage where the Tableau dashboard can be viewed directly in a browser.

9. Project Demonstration

The demonstration video explains dataset usage, data preparation, dashboard creation, story presentation, and web integration. Recommended video length is 5–10 minutes.

10. Conclusion

The Sales Analysis Dashboard demonstrates how data visualization can convert raw data into business insights. Tableau enables interactive analysis of sales trends, profitability, and product performance to support strategic decisions.